CANDIDATE NAME: ______________________________________ INDEX NUMBER _________

CG _________

SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2018

H1 BIOLOGY **8876/01**
Paper 1 Multiple Choice

Friday
21 September 2018

Additional Materials: Multiple Choice Answer Sheet **1 hour**

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CG in the spaces at the top of this page.

On the Multiple Choice Answer Sheet, write your name, subject title, test name and CG. For your index number, write your full NRIC number. Shade the corresponding lozenges on the Answer Sheet according to the instructions given by the invigilators.

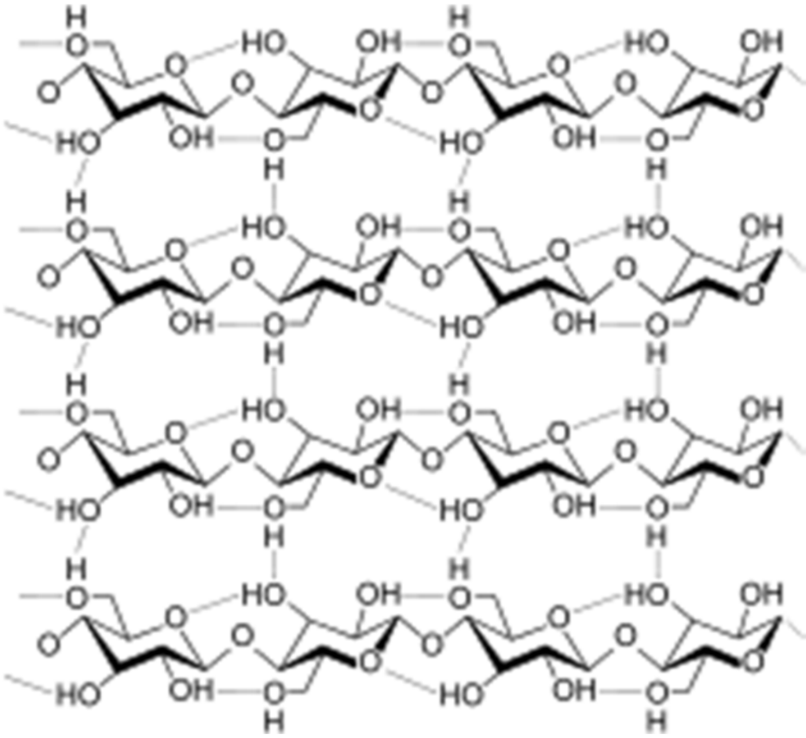
There are **thirty questions** on this paper. Answer **all** questions. For each question, there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

AT THE END OF THE EXAMINATION, SUBMIT BOTH THE MULTIPLE CHOICE ANSWER SHEET AND THE QUESTION PAPER.

This question paper consists of **18** printed pages including this cover page.

1. The following biomolecule is



A not a protein because of the presence of phenyl groups.

B a protein because of the presence of peptide linkages.

C not a carbohydrate because of the presence of regular repeated folding.

D a carbohydrate because of the presence of glycosidic linkages.

2. Some foods contain ‘hydrogenated vegetable oils’. These are unsaturated fats that have been converted to saturated fats.

Which property of the fats will have changed?

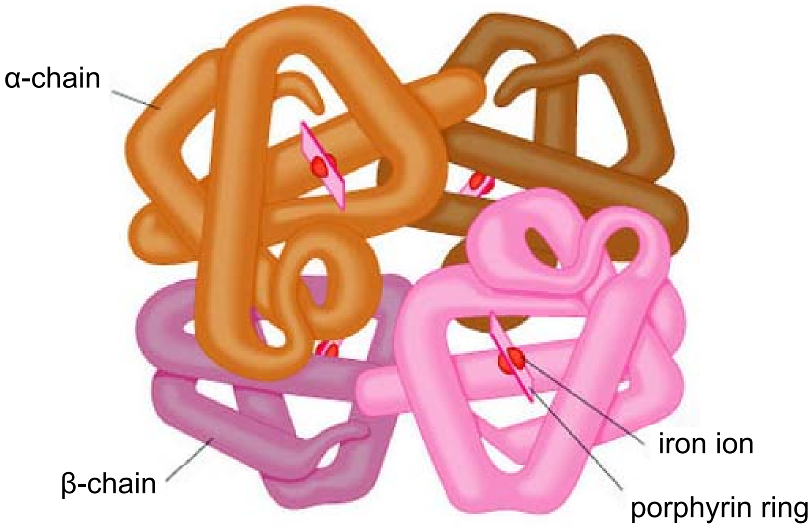
A Their hydrocarbon chains will fit together more closely.

B Their solubility in water will increase.

C They will have more double bonds in their molecules.

D They will remain liquid at room temperature.

3. A simplified representation of a haemoglobin molecule is shown below.



Several interacting pairs are listed below.

1 α-chain and β-chain
2 α-chain and porphyrin ring
3 iron ion and oxygen gas

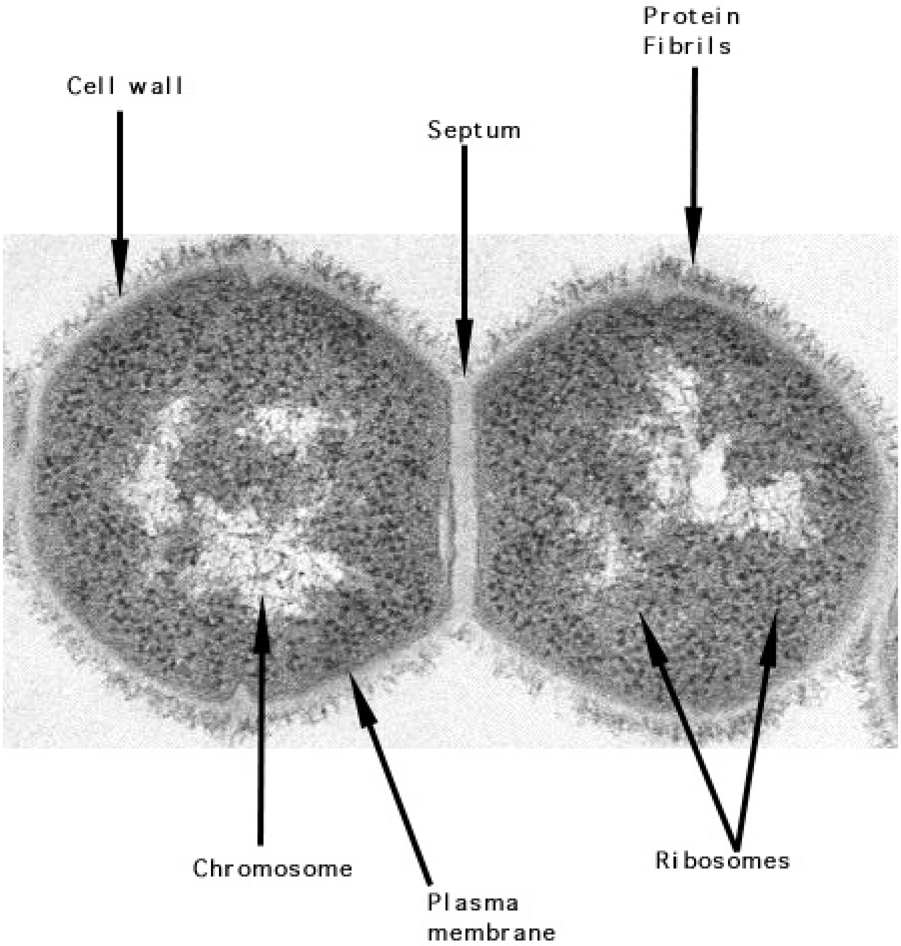
Which of the above pairs interact via weak interactions?

A 1 and 2 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 3

4. Which of the following does **not** describe a feature of an organelle that contributes to its respective function?

A The interior of a lysosome is acidic.
B The ribosome has rRNA and protein components.
C The Golgi body consists of membranous sacs that are independent.
D The smooth endoplasmic reticulum is not continuous with the nuclear membrane.

5. The following is an electron micrograph of a living cell that is dividing.



The following are some possible statements about the daughter cells:

1 These are a pair of plant cells as the septum represents the cell plate of the dividing cell.
2 These are a pair of bacterial cells as they have chromosomes and ribosomes.
3 These cannot be bacterial cells as they have a cell wall.
4 These are a pair of animal cells as they have proteins on their outer surface.
5 These cannot be animal cells as the ribosomes are located alongside the chromosomes.
6 These cannot be bacterial cells as circular DNA is not evident.

Which of the statements about these cells are correct?

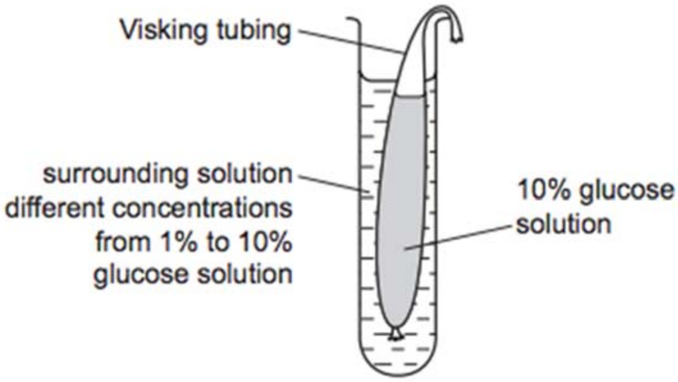
A 1 and 3

B 1 and 6

C 3 and 4

D 2 and 5

6. The diagram shows apparatus set up to investigate the effect of changing the concentration of glucose in the surrounding solution on the movement of molecules through a selectively permeable membrane (Visking tubing) in 15 minutes.

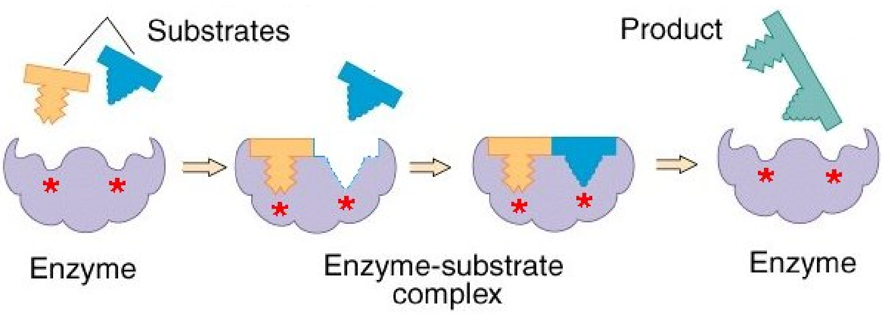


As the concentration of glucose solution in the surrounding solution increases, which statements are correct?

1 Net diffusion of water increases.
2 Glucose molecules reach an equilibrium quicker.
3 There is less change in the volume of surrounding solution.
4 Net diffusion of glucose increases.

A 1 and 3 only
B 2 and 3 only
C 1, 2 and 4 only
D 1, 2, 3 and 4

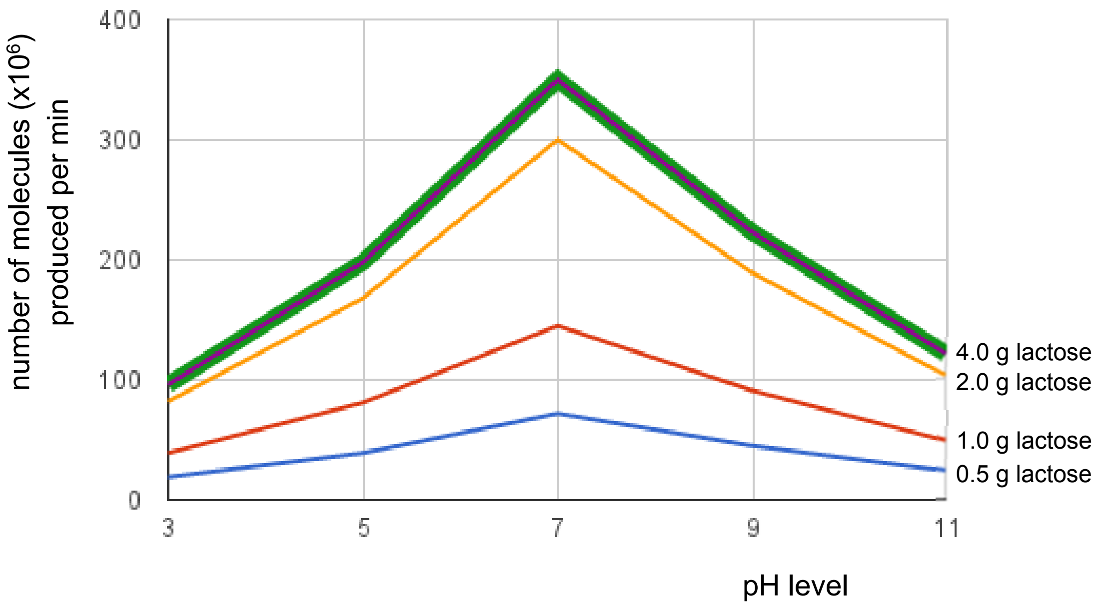
7. The figure below shows the mode of action of a particular enzyme.



Which of the following is **only** true for the mode of enzyme action shown above?

A A few amino acids give the active site a specific conformation.
B Both contact and catalytic residues interact with the substrate.
C The enzyme changes shape in the presence of the substrates.
D The substrate molecules are complementary to the active site.

8. The following is a graph showing the number of molecules of product produced from the digestion of lactose at different pH levels.



Which of the following is **not** a valid conclusion from this graph?

A Enzyme molecules are fully denatured at extreme pH levels.

B The reaction rate is likely to level off past a certain substrate concentration.

C pH level of 7 is the optimal pH for this enzyme.

D Increasing substrate concentration increases the rate of enzyme reaction.

9. The following are some statements related to cell and nuclear division in humans.

1 A diploid cell that results from mitosis will have the same alleles at the same loci.

2 The cell size increases significantly during cytokinesis.

3 Meiosis ensures that a fertilised cell has the diploid number of chromosomes.

4 The cell cycle of a specific cell includes either meiotic or mitotic division but never both at the same time.

Which pair of the above statements is **correct**?

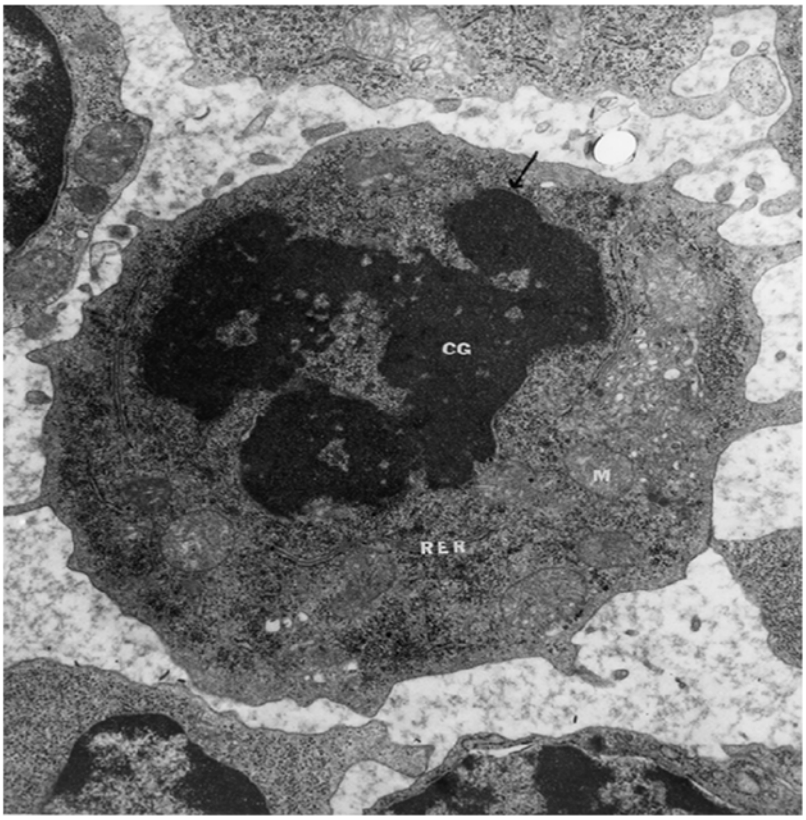
A 1 and 2 only

B 1 and 4 only

C 2 and 3 only

D 3 and 4 only

10. The following depicted cell is undergoing cell division.



(CG = chromatin granules, RER = rough endoplasmic reticulum, M = mitochondrion)

Which of the following is/are correct observations of the cell above that indicate(s) that it is undergoing division?

1 The nuclear membrane has been broken down.
2 Opposite poles have been established.
3 Rod-like chromosomes can be observed.

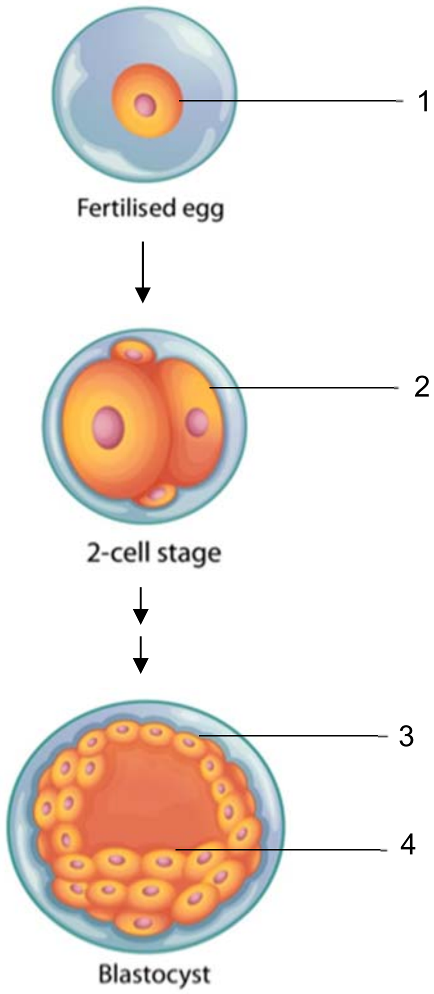
A 1 only

B 1 and 2 only

C 1 and 3 only

D None of the above

11. The figure below shows some stages of embryonic development. Four different cells are labelled 1, 2, 3 and 4 as shown.



Which of the following statements is incorrect?

A Cells 1 and 2 are totipotent.

B Cells 3 and 4 are pluripotent.

C Cell 3 is a result of mitoses involving Cell 1.

D Cells 2 and 4 can give rise to adult stem cells.

12. Scientists have made a nucleic acid (HNA) that has a sugar with the same number of carbon atoms as glucose instead of deoxyribose. Although genetic information can be stored by HNA, naturally occurring DNA polymerase cannot replicate HNA.

Which statements could explain why naturally occurring DNA polymerase cannot replicate HNA?

1 DNA polymerase cannot form bonds between the sugars of two HNA nucleotides.

2 DNA polymerase cannot form hydrogen bonds between two HNA nucleotides.

3 HNA nucleotides do not fit into the active site of DNA polymerase.

4 The shape of an HNA nucleotide is slightly larger than that of a DNA nucleotide.

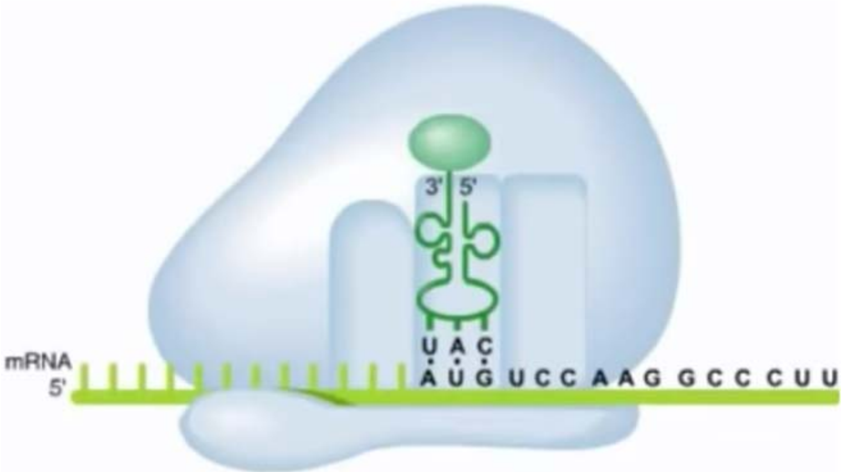
A 1, 2, 3 and 4

B 1 and 4 only

C 2 and 3 only

D 3 and 4 only

13. The figure below shows a stage in translation.



All of the following are events that occur after this stage **except** the

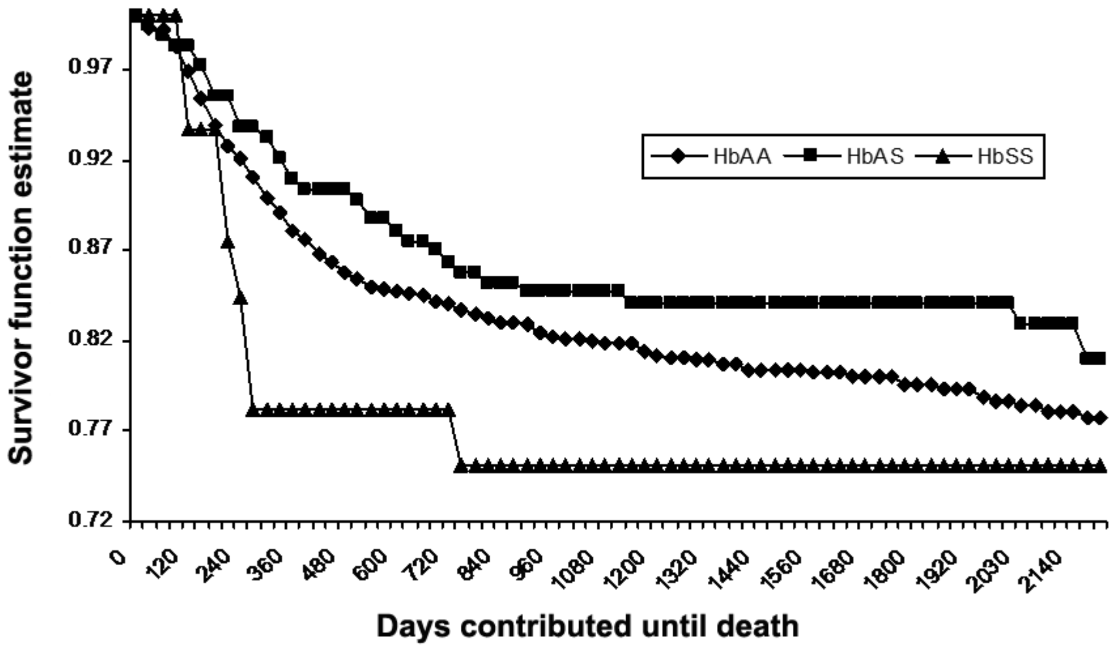
A addition of a poly(A) tail.

B folding of the polypeptide chain.

C use of GTP as an energy source.

D formation of peptide and hydrogen bonds.

14. The graph below shows the survival curves of children of the Luo ethnic group, which lives in an area of Kenya with widespread malaria.



(HbAS: Heterozygous sickle-cell haemoglobin; HbAA: normal haemoglobin; HbSS: homozygous sickle-cell haemoglobin.)

Which of the following is a valid interpretation of the data above?

A HbSS children have a higher chance of dying from sickle-cell anaemia than HbAA children.

B HbAS and HbSS children are selected against by the malarial parasite compared to HbAA children.

C HbAS children have a higher chance of dying due to susceptibility to both malaria and sickle-cell anaemia.

D HbAA and HbAS children show the same phenotype at all times and hence have similar survival rates.

15. The six statements are about the genes involved in the synthesis of particular glycoproteins referred to as human blood group antigens.

1 There are two genes, one carried on chromosome 19 and the other on chromosome 9.

2 The gene on chromosome 19 codes for an enzyme used for the synthesis of antigen H on the cell surface membrane of red blood cells.

3 The gene on chromosome 9 has variants that each codes for one of three different enzymes, two active and one inactive.

4 The active enzymes modify antigen H to produce different antigens.

5 If both active enzymes are present, red blood cells have two different modified antigens.

6 If only the genetic code for inactive enzyme is present, red blood cells have unmodified antigen H.

Which row links each genetic term to one or more statements that include an example of the genetic term?

	allele	codominant	locus	recessive	phenotype
A	1	4 and 5	2	3	5
B	1	4	2 and 3	2	3, 4 and 5
C	3	1	3	6	6
D	3	5	1, 2 and 3	6	5 and 6

16. In the magpie moth *Abraxas sp.*, the female is the heterogametic sex and the gene for wing colour is sex-linked.

In a cross between a normal coloured male and a pale coloured female, the F1 offspring consisted of all normal coloured individuals with the two sexes in equal proportions.

Which ratio would be obtained in the F2 generation produced from the F1 generation?

A Normal coloured males to normal females 1:1

B Normal coloured males and females to pale females 3:1

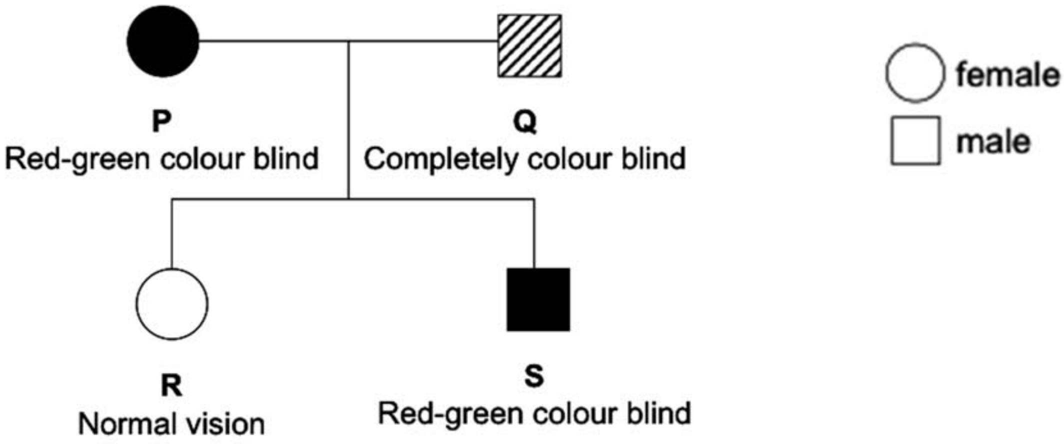
C Normal coloured males and females to pale males and females 1:1

D Normal coloured males to pale coloured females 1:1

17. Red-green colour blindness is controlled by a gene on the X chromosome. The allele for colour blindness, **g**, is recessive to the allele for normal colour vision, **G**.

Complete colour blindness is controlled by a different gene which is not on the X chromosome. The allele for the development of normal cones (pigment cells in the retinal layer of the eye), **B**, is dominant to the allele for no cone development, **b**.

The figure below shows the phenotypes of members of a different family in which both types of colour blindness occur.

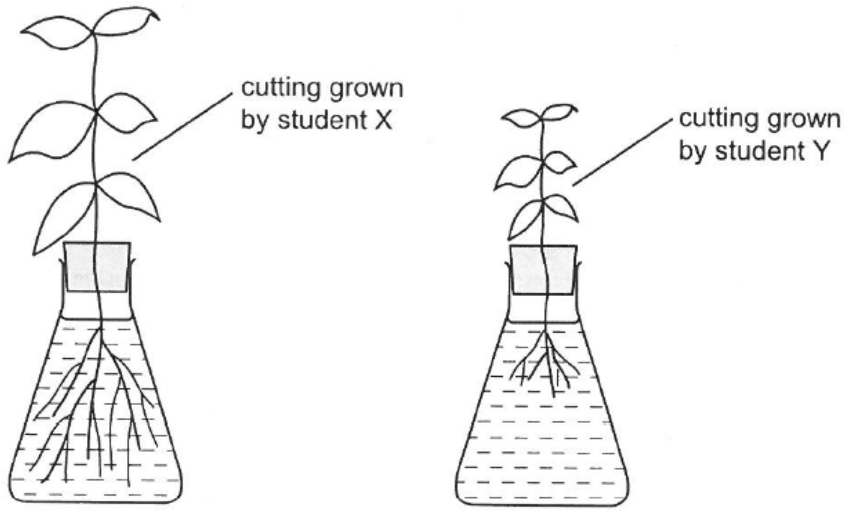


Which of the following are possible genotypes for individuals P, Q and S?

	P	Q	S
A	BBX^GX^G	bbX^GY	BBX^gY
B	BbX^GX^g	BbX^gY	BbX^GY
C	BBX^gX^g	bbX^GY	BbX^gY
D	BbX^GX^g	BbX^gY	BBX^GY

18. A student, X, looked after a plant. Another student, Y, looked after another plant of the same species. Each student followed the same instructions to set up apparatus to take cuttings from their plant and grow the cuttings next to the plant from which the cutting had been taken.

The diagrams show the results after one week.



Which factors could have caused the different appearance of the two cuttings?

1 phenotypic variation due to the environment
2 genetically different cuttings
3 mutation due to environment
4 genotypic variation due to environment

A 1 and 2 **B** 2 and 3 **C** 2 and 4 **D** 1, 3 and 4

19. Which stages of aerobic respiration in eukaryotes have the correct products?

	ATP	CO_2	FAD	NAD	reduced NAD
1 glycolysis	✓	×	×	×	✓
2 oxidative phosphorylation	✓	×	✓	✓	×
3 Krebs cycle	✓	✓	✓	×	✓
4 link reaction	✓	✓	×	×	✓

Key: ✓ = product × = not a product

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

20. In living cells, 2,4 dinitrophenol acts as a proton ionophore, an agent that can shuttle protons across biological membranes.

Which of the following is **not** a possible consequence of the introduction of 2,4 dinitrophenol into an animal cell?

A Less oxygen is taken up by the cell.

B The proton gradient across the inner mitochondrial membrane is dissipated.

C The rate of glycolysis in the cell will increase.

D The rate of Krebs cycle in the cell will increase.

21. There are two main forms of respiration, aerobic respiration and anaerobic respiration.

Which of the following only occurs during one form of respiration, but not the other?

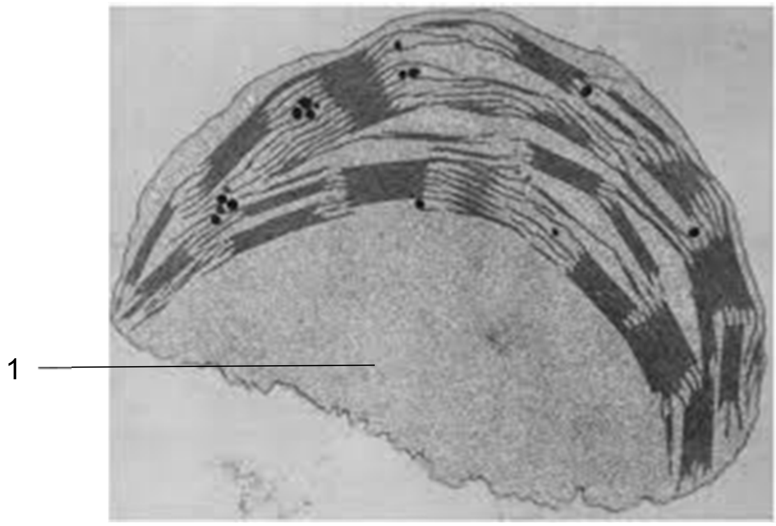
A Use of mitochondrial transport proteins

B Regeneration of hydrogen carriers

C Release of energy

D Substrate-level phosphorylation

22. An electron micrograph of a chloroplast is shown below, with region 1 marked out.



Which of the following does **not** occur in region 1?

A Synthesis of carbohydrates

B Storage of carbohydrates

C Synthesis of proteins

D Trapping of light

23. ...1… and …2… are reactants of the light-dependent stage of photosynthesis, but …3… is not a reactant of this stage.

	1	2	3
A	water	carbon dioxide	oxygen
B	NADP	carbon dioxide	water
C	water	NADP	ATP
D	NADP	ADP	water

24. Which of the following best explains why the population is the smallest unit that can evolve?

A There must be a group of individuals which have different phenotypes in order for selection to occur.

B Mutations must occur in different individuals in order for selection to occur.

C Individuals must be capable of producing fertile offspring in order for selection to occur.

D Individuals must be able to interact with their biotic and abiotic environment in order for selection to occur.

25. Deer mice are small, ground dwelling rodents. They normally have dark coats.

Some paler coated mice have been observed in an area where new sand hills were formed between 8000 and 15 000 years ago. The sand hills are sparsely covered with scrubby plants.

Studies have shown that the change in coat colour was due to a mutation in a gene, causing a single amino acid deletion from the protein for the coat colour pigment. This mutation seems to be spreading through the population.

Which statement most fully explains how the evolution of this species at this site has occurred?

A Better camouflage increases the chance of survival by reducing predation.

B Deer mice which live longer usually leave more offspring since they have more reproductive opportunities.

C The occurrence of the mutation provided new variation on which natural selection can act.

D The paler colour gives better camouflage against the background of the new sand hills.

26. New research conducted by evolutionary biologists worldwide paints cities as evolutionary "change agents", says a trio of biologists from the University of Toronto Mississauga (UTM). A compilation of 15 new research papers, published as a special issue of *Proceedings of the Royal Society B: Biological Sciences*, confirms that cities frequently alter evolution by natural selection.

The following statements are possible ways in which cities could alter evolution by natural selection.

1 Cities are generally warmer than natural areas and thus organisms adapted to higher temperatures would be selected for within cities.

2 Cities release large amounts of environmental pollutants and thus organisms with greater resistance to common pollutants would be selected for in areas within cities.

3 Cities provides additional food sources for many organisms and thus organisms that are adapted to feed on a wider range of food types would be selected for within cities.

4 Cities have a large human population so organisms that are better able to interact with humans will be selected for within the cities.

Which statements are potentially correct?

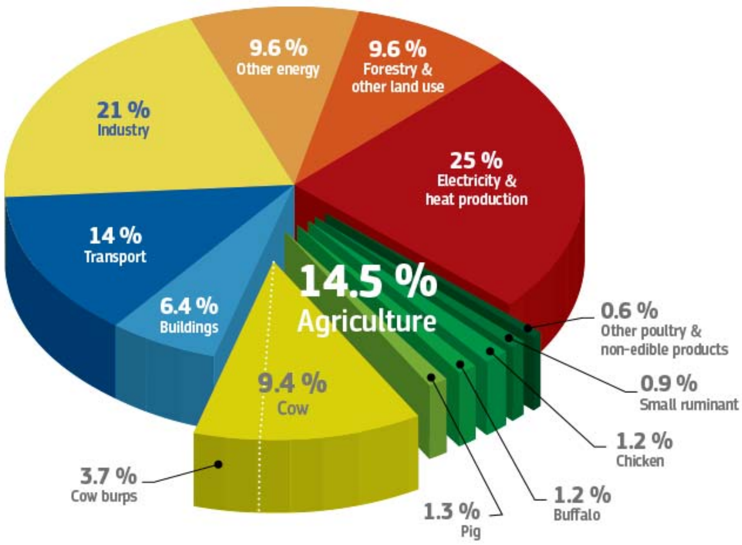
A 1 and 2 only

B 2 and 4 only

C 1, 2 and 3 only

D 1, 2, 3 and 4

27. The data below on greenhouse gas emissions by economic sector is reported by the Intergovernmental Panel on Climate Change and Food and Agriculture Organisation of the United Nations in 2015.



Based on the figure, what proportion of greenhouse gas emissions can be attributed to the use of fossil fuels?

A 25% **B** 46% **C** 76% **D** 85.6%

28. It has been discovered that deep corals, which are found at ocean depths below the reach of sunlight, is also affected by climate change.

Which of the following statements is a valid explanation as to why this is so?

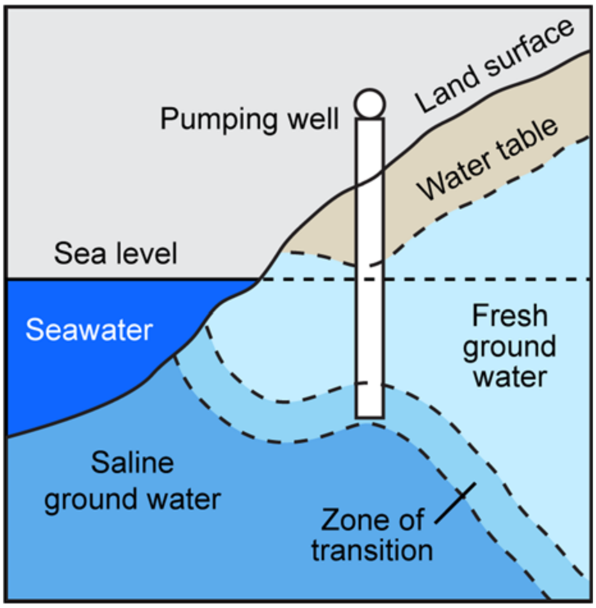
A The rate of photosynthesis at the deeper waters inhabited by the deep coral species is inhibited by a lack of carbon dioxide.

B Deep water coral species are adapted to lower temperatures and are unable to migrate to shallow waters which have a higher water temperature.

C The warming of surface water temperatures due to global warming has led to even the deeper waters heating up beyond the natural range of deep water coral species.

D The rising sea levels globally have led to deep corals being unable to receive sunlight for use by its symbiotic algae.

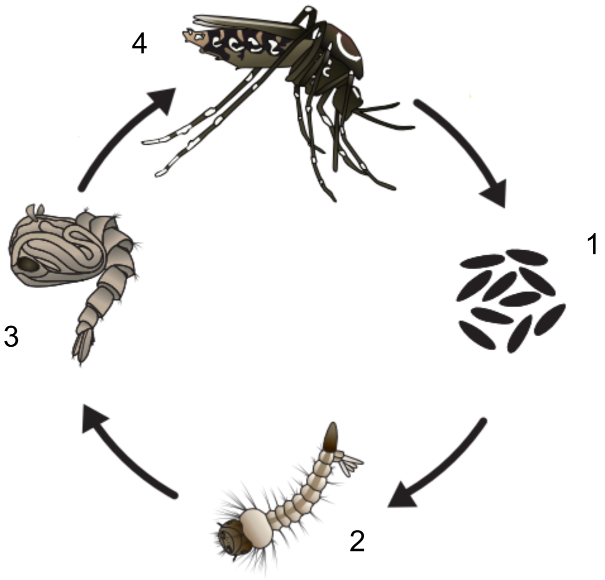
29. The figure below shows the distribution of water resources along a coast.



Which of the following reflects the correct changes in volume that occur with global warming?

	Seawater	Saline ground water	Fresh ground water
A	increase	decrease	decrease
B	increase	increase	decrease
C	decrease	decrease	increase
D	decrease	increase	increase

30. The life-cycle of *Aedes aegypti* is shown below.



Which is a correct statement related to the information above?

- **A** Stage 1 can withstand prolonged periods of dry conditions.
- **B** Only two out of the four stages above occur in water.
- **C** Stage 3 is infected by the dengue virus before stage 4 infects human hosts.
- **D** Both males and females of stage 4 can act as vectors for the dengue virus.

END OF PAPER